ASSIGNMENT TIME-SERIES

Time-Series Forecast of PM2.5 Concentration in Delhi

Using SARIMA and Log-SARIMAX Models

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Abstract

This study examines short-term forecasting of daily PM2.5 concentration in Delhi using classical time-series and SARIMA-family models [3, 8]. The analysis uses the public *Air Quality Data in India* dataset and restricts the sample to Delhi from 1 January 2015 to 1 July 2020 [1]. A chronological train-test split is applied, with observations from 2015–2019 used for training and observations from 1 January 2020 to 1 July 2020 reserved for out-of-sample evaluation. The modeling workflow compares simple baseline forecasts, Holt-Winters ETS, SARIMA, and a Log-SARIMAX extension with selected exogenous pollutant variables.

The empirical results show that seasonal structure is important for PM2.5 forecasting. Naive forecasting performs poorly, while seasonal naive and ETS baselines improve accuracy by capturing recurring pollution patterns. The selected SARIMA$(3,1,1)(0,1,1)_7$ model captures short-run persistence and weekly seasonal dependence, and residual Ljung-Box diagnostics indicate no significant remaining autocorrelation at weekly lags. However, static SARIMA forecasts over the full holdout period perform weakly because errors accumulate when the model is not updated. Walk-forward validation produces much better SARIMA accuracy, showing that the model is more useful in an operational daily-updating setting. Log-SARIMAX further improves short-horizon forecasting by combining log transformation, seasonal time-series structure, and exogenous pollutant information from Benzene, NOx, and NH3. The study concludes that walk-forward SARIMA is a transparent univariate benchmark, while Log-SARIMAX is preferable when reliable exogenous pollutant inputs are available.

Keywords: PM2.5 forecasting; Delhi air pollution; SARIMA; Log-SARIMAX; time-series forecasting; walk-forward validation; air quality monitoring.

Contents

Introduction

Delhi is one of the world's most polluted large urban areas and experiences repeated episodes of severe particulate pollution, especially during late autumn and winter [1]. During these periods, emissions from transport, construction, industry, household fuel use, regional crop-residue burning, and unfavorable meteorological conditions combine to produce high concentrations of fine particulate matter. Among air pollutants, PM2.5 is particularly important because its small particle size allows it to penetrate deeply into the respiratory system and because sustained exposure is associated with substantial public-health risks. Forecasting daily PM2.5 concentration is therefore useful for early warning systems, school and workplace advisories, hospital preparedness, and short-run pollution-control decisions.

The central problem in this project is to predict daily PM2.5 concentration in Delhi using historical time-series patterns. The target series contains persistence, seasonality, sharp pollution spikes, and periods of changing volatility. These features make forecasting challenging: a useful model must learn regular temporal dependence while also remaining accurate during abrupt pollution episodes. The empirical question is therefore not only whether past PM2.5 values contain predictive information, but also whether seasonal dynamics and additional pollutant variables can improve short-term forecast accuracy.

The objective of the study is to compare several forecasting approaches on the same out-of-sample holdout period. Simple baseline models are first used as reference benchmarks for judging whether more complex methods add value, including naive, seasonal naive, and ETS forecasts. A SARIMA model is then estimated to capture autoregressive, moving-average, differencing, and seasonal patterns in the PM2.5 series. Finally, a Log-SARIMAX specification is considered to stabilize variance through a logarithmic transformation and to incorporate selected exogenous pollutant variables. By evaluating the baseline models, SARIMA, and Log-SARIMAX under the same train-test split, the project provides a transparent comparison of model accuracy and identifies which approach is most suitable for short-term PM2.5 forecasting in Delhi.

One | Literature Review

Time-series forecasting has been widely used in air-pollution research because pollutant concentrations usually display temporal dependence, short-run persistence, and recurring seasonal patterns. For daily PM2.5, today's concentration is often related to recent observations because emissions, atmospheric stability, and pollutant accumulation do not change independently from one day to the next. Classical autoregressive integrated moving average models are therefore useful as transparent forecasting tools [3, 8]. ARIMA models represent the current value of a series through past values, differencing terms, and past forecast errors, while SARIMA extends this structure by adding seasonal autoregressive and moving-average components. In pollution applications, this seasonal extension is important because particulate matter often follows weekly, monthly, or annual cycles driven by weather, human activity, and seasonal emission sources. Although ARIMA and SARIMA are simpler than many machine-learning methods, they remain strong baseline models because they directly capture persistence and seasonality in a univariate time series [3, 8].

Another common direction in forecasting research is ensemble forecasting [2, 4]. Instead of relying on a single model, an ensemble combines predictions from several models in order to reduce the effect of individual model error. The basic idea is that different models may capture different aspects of the same data-generating process: one model may follow short-run persistence well, another may respond better to seasonal movement, and another may handle nonlinear behavior. Averaging or weighting their forecasts can sometimes produce more stable predictions than any single component model [2, 4]. However, this study does not implement a full ensemble pipeline. The analysis focuses on separate model comparison rather than formal forecast combination, so ensemble forecasting is discussed only as a related approach and not as part of the empirical workflow.

Recent PM2.5 forecasting studies commonly compare several modeling strategies instead of evaluating one method in isolation [2, 5, 6, 9, 10]. Statistical models such as ARIMA, SARIMA, and exponential smoothing are often used as interpretable benchmarks [3, 8]. Transformed models are also considered when the original pollutant series has skewness, changing variance, or extreme spikes; applying a logarithmic transformation can make the series more stable and may improve forecast performance. In addition, hybrid approaches that combine statistical models with machine-learning or deep-learning components have become increasingly common, especially when researchers want to capture both linear temporal structure and nonlinear pollutant dynamics [2, 4, 5, 10, 12]. This project follows the comparative spirit of this literature by evaluating baseline forecasts, ETS, SARIMA, and Log-SARIMAX on the same holdout period [3, 8, 9]. The goal is not to build the most complex possible forecasting system, but to determine whether seasonal modeling, variance stabilization, and selected exogenous pollutant information improve daily PM2.5 forecasts beyond simpler benchmarks.

Two | Data and Preprocessing

This study uses the public *Air Quality Data in India* dataset distributed on Kaggle by Rohan Rao [1]. The dataset is a curated release of city-level air-quality observations originally sourced from the Central Pollution Control Board (CPCB), India's national pollution-control authority. The full Kaggle file, `city_day.csv`, contains daily observations for multiple Indian cities, but this project restricts the analysis to Delhi in order to construct a single-city time series suitable for SARIMA forecasting. This restriction is methodologically important because SARIMA assumes a temporally ordered sequence from one data-generating process; combining cities would introduce cross-sectional heterogeneity that is not part of the present modeling objective.

The resulting Delhi subset is used as the empirical foundation for the report. The response variable is daily PM2.5 concentration, while the remaining pollutants and AQI variables provide contextual information for understanding Delhi's broader air-quality conditions. This chapter documents the source, coverage, preprocessing decisions, and target-series characteristics that motivate the forecasting strategy.

DATASET SCOPE

After filtering the original source to Delhi, the working dataset contains 2,009 daily observations from 1 January 2015 to 1 July 2020. The dataset confirms that each row corresponds to one date in Delhi and that the columns include `City`, `Date`, `PM2.5`, `PM10`, `NO`, `NO2`, `NOx`, `NH3`, `CO`, `SO2`, `O3`, `Benzene`, `Toluene`, `AQI`, and `AQI_Bucket`. These variables capture both particulate pollution and gaseous pollutants associated with traffic, combustion, industrial activity, secondary atmospheric reactions, and volatile organic compounds.

The target variable for forecasting is `PM2.5`, measured as fine particulate matter with aerodynamic diameter less than or equal to 2.5 micrometers. PM2.5 is selected because it is one of the most policy-relevant air-pollution indicators and because its daily sequence is suitable for short-term time-series forecasting. Although the dataset includes several explanatory pollutants, the baseline model is intentionally univariate so that historical PM2.5 behavior can first be evaluated as a transparent forecasting benchmark.

DATA PREPARATION AND TRAIN-TEST SPLIT

The preprocessing stage converts the `Date` column from string format to a datetime object, orders the observations chronologically, and sets the date as the time index for modeling. Because the analysis is restricted to Delhi, the city identifier is removed from the modeling dataset. The target series is daily PM2.5 concentration, denoted by y_t.

The data quality audit shows that all variables used in the working modeling dataset are complete after preprocessing and that there are zero duplicate rows. This is important for time-series forecasting because missing values, duplicated dates, and inconsistent ordering can distort autocorrelation patterns and bias the estimation of autoregressive and moving-average terms.

Temporal information is extracted from the date variable by constructing calendar features such as year, quarter, month, day of week, and week of year. These variables are used for exploratory analysis rather than directly imposed in the baseline SARIMA model. They help identify recurring seasonal patterns, compare weekday and weekend pollution levels, and locate periods in which extreme PM2.5 observations occur.

To preserve the temporal ordering of the series, the sample is divided using a chronological train-test split rather than a random partition. The first five years of observations, from 1 January 2015 to 31 December 2019, are assigned to the training set and used for model identification, parameter estimation, and diagnostic checking. The remaining observations, from 1 January 2020 to 1 July 2020, are reserved as the out-of-sample test set for evaluating forecast accuracy. This gives a holdout period of roughly six months, which is long enough to compare forecast behavior across different pollution conditions while remaining fully later than the estimation window. This design prevents data leakage because no information from the 2020 evaluation period is used during model estimation. The forecasting task is therefore defined as predicting future PM2.5 values $\hat{y}_{t+h}$ using only observations available before the forecast origin.

EXPLORATORY DATA ANALYSIS AND STATIONARITY

The exploratory data analysis focuses on the target variable, daily PM2.5 concentration. This focus is important because the baseline SARIMA model is univariate: it forecasts future PM2.5 values from the past behavior of PM2.5 itself. Other pollutants are considered later only in the Log-SARIMAX extension, while this subsection establishes whether the target series has the distributional, seasonal, and dependence patterns required for time-series modeling.

The PM2.5 series is highly variable and right-skewed. Across the 2,009 daily observations, PM2.5 ranges from 10.24 to 685.36 μg/m^3, with a mean of 117.10 μg/m^3, a median of 94.49 μg/m^3, and a standard deviation of 82.93 μg/m^3. The mean is higher than the median, and the distribution has positive skewness of 1.705 and kurtosis of 4.504. This indicates that most observations are moderate relative to the most extreme values, but a small number of severe pollution episodes create a long upper tail. These spikes are important because they can substantially increase forecast error even when the model captures average daily dynamics reasonably well.

The time-based summaries show a strong seasonal structure in PM2.5. Quarterly averages indicate that PM2.5 is highest in Q4 at 195.29 μg/m^3, followed by Q1 at 142.36 μg/m^3, while Q3 is the cleanest period at 48.12 μg/m^3. The monthly pattern is consistent with this result: concentrations rise sharply in late autumn and winter, especially around November and December, and fall during the monsoon and summer periods. In contrast, the weekday-weekend difference is small, with average PM2.5 of 117.39 μg/m^3 on weekdays and 116.38 μg/m^3 on weekends. Thus, broad seasonal forces appear more important than simple day-of-week effects for this dataset.

Extreme-value analysis reinforces the seasonal interpretation. The five highest PM2.5 observations occur in early November, including 685.36 μg/m^3 on 6 November 2016 and 639.19 μg/m^3 on 8 November 2017. The lowest observations occur mainly in July, August, and September, including 10.24 μg/m^3 on 17 August 2019 and 10.88 μg/m^3 on 30 July 2017. These extremes show that the target series contains both regular seasonal movement and abrupt high-pollution shocks. A forecasting model must therefore capture persistence and seasonality while remaining robust to short-run volatility.

Correlation and multicollinearity diagnostics are then used to screen possible exogenous variables for the Log-SARIMAX extension. PM2.5 is most strongly correlated with AQI ($r = 0.882$), PM10 ($r = 0.839$), Benzene ($r = 0.698$), NO ($r = 0.668$), NO2 ($r = 0.647$), NH3 ($r = 0.584$), and NOx ($r = 0.488$). However, high correlation with PM2.5 is not sufficient for selecting regressors because some predictors are highly redundant with each other. VIF analysis shows severe multicollinearity for AQI (23.60), NO2 (23.34), and PM10 (20.74), while NOx (8.72), NH3 (8.24), SO2 (8.10), O3 (7.93), NO (7.59), and Toluene (4.00) fall below the conventional VIF threshold of 10. Therefore, the exogenous feature set is kept selective: NOx and NH3 are retained because they satisfy the VIF criterion, while Benzene is also included because it has a strong association with PM2.5 and only slightly exceeds the VIF threshold. This gives the Log-SARIMAX model additional pollutant information without including all highly collinear variables.

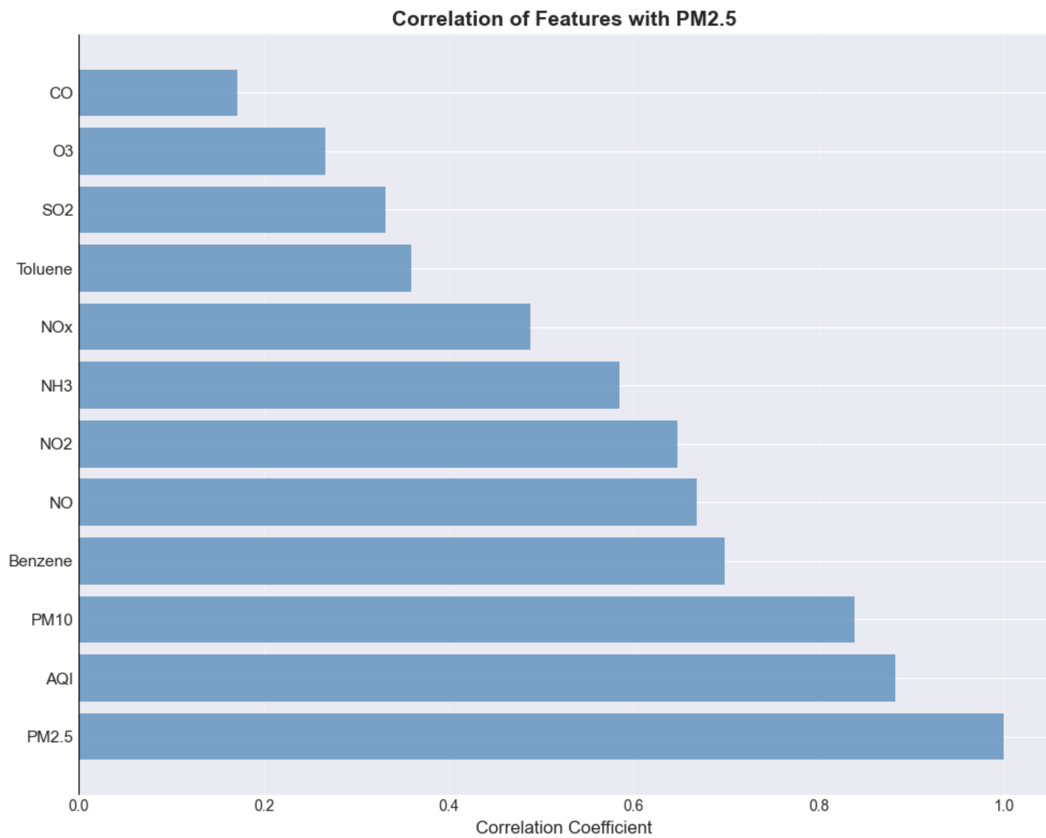

Figure 1: Correlation matrix of PM2.5 and related air-pollution variables.

Before estimating forecasting models, the PM2.5 training series is examined through rolling statistics, seasonal decomposition, stationarity tests, and autocorrelation diagnostics. These procedures connect the exploratory findings to the statistical assumptions of SARIMA modeling. They help determine whether the series contains a changing mean, changing variance, seasonal structure, persistence, and irregular shocks that should be addressed before model estimation.

The seven-day rolling mean and rolling standard deviation are used to evaluate local changes in level and volatility. The rolling mean fluctuates substantially over time and displays repeated peaks during high-pollution winter periods and lower values during cleaner periods. The rolling standard deviation also changes over time: volatility is higher during polluted episodes and lower during cleaner periods. This suggests heteroscedasticity in the original PM2.5 series and provides a practical motivation for considering a logarithmic transformation in the Log-SARIMAX extension.

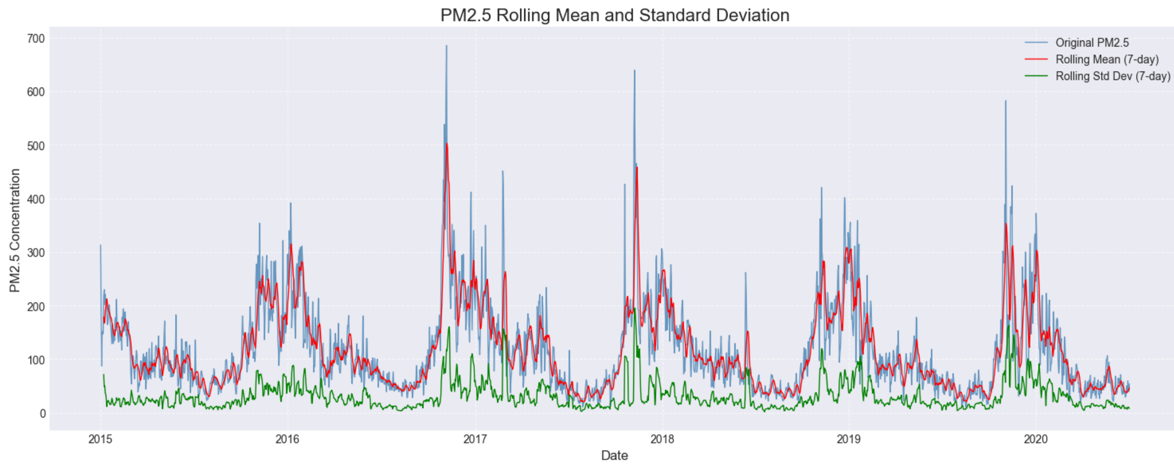

Figure 2: Seven-day rolling mean and rolling standard deviation of Delhi PM2.5.

Seasonal decomposition is then applied using an additive representation:

$$y_t = T_t + S_t + R_t,$$

where T_t is the trend component, S_t is the seasonal component, and R_t is the irregular residual component. Because the data are daily and the exploratory summaries show a clear winter–summer cycle, the decomposition uses an annual seasonal period of $s = 365$. The annual component confirms that PM2.5 rises during winter months and declines during summer and monsoon periods, with seasonal fluctuations roughly between -100 and $+220$ $\mu g/m^3$. The trend component changes gradually and shows a mild downward movement after 2017, while the residuals are mostly centered around zero but still contain large positive spikes. These remaining spikes suggest exceptional pollution shocks that are not fully explained by regular annual seasonality. Thus, annual seasonality is important, but SARIMA with $s = 365$ is computationally expensive for daily data spanning multiple years.

Stationarity is assessed using both the Augmented Dickey-Fuller (ADF) and KPSS tests because the two tests use opposite null hypotheses. The ADF test evaluates the null hypothesis of a unit root, whereas the KPSS test evaluates the null hypothesis of stationarity. For the original PM2.5 series, the ADF test rejects the unit-root null with a p-value of 0.0015, while the KPSS test fails to reject stationarity with a p-value of 0.1000. Taken together, these results suggest that the original series is statistically stationary in the stochastic sense, even though the rolling diagnostics and decomposition reveal strong seasonal fluctuation and time-varying volatility. Therefore, non-seasonal differencing is not automatically required, and $d = 0$ remains a valid candidate during model selection.

Autocorrelation diagnostics are used to guide the autoregressive and moving-average orders. The autocorrelation function (ACF) measures the correlation between y_t and y_{t-k} at lag k, while the partial autocorrelation function (PACF) measures the direct relationship between y_t and y_{t-k} after controlling for intermediate lags. For the original PM2.5 series, the ACF declines slowly and remains positive across many lags, indicating strong persistence. The PACF shows a very large first-lag spike, approximately 0.87, followed by much smaller values near the confidence threshold. This pattern is consistent with a low-order autoregressive component, with AR(1) providing a natural starting point for candidate specifications.

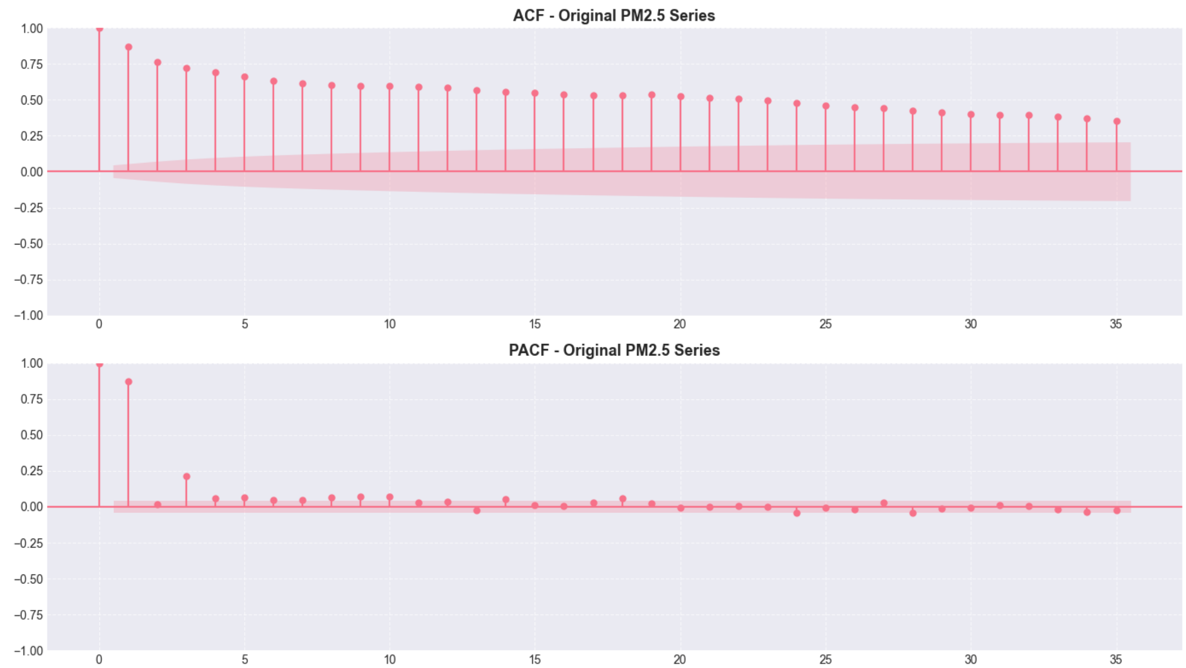

Figure 3: ACF and PACF of the original PM2.5 training series.

Differencing is nevertheless examined as a robustness step. Ordinary differencing is defined as

$$\nabla y_t = y_t - y_{t-1},$$

while seasonal differencing is defined as

$$\nabla_s y_t = y_t - y_{t-s}.$$

The first-differenced series fluctuates around a mean close to zero and removes much of the long-run dependence, but its ACF contains negative early-lag spikes, suggesting possible over-differencing. By contrast, seasonal differencing with $s = 7$ more directly targets weekly dependence: the ACF declines toward zero after the seasonal adjustment, while the PACF still shows meaningful signals at lag 1 and at seasonal lags such as 8, 15, and 22. This implies that seasonal differencing helps reduce recurring dependence, but autoregressive structure remains relevant.

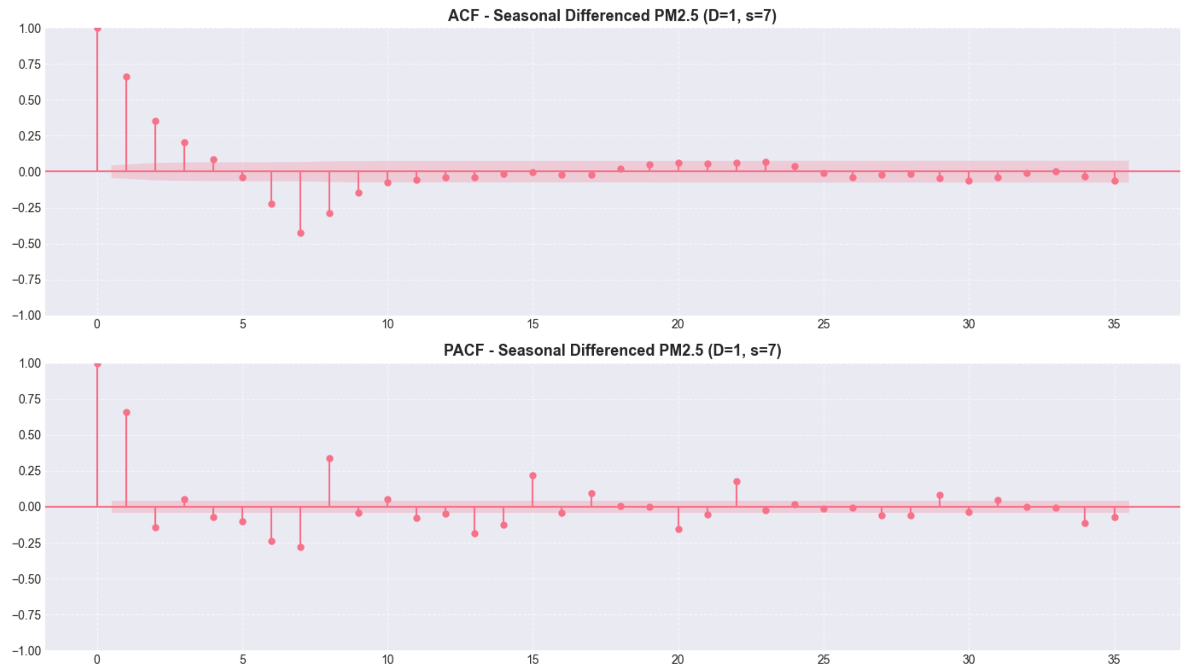

Figure 4: ACF and PACF after weekly seasonal differencing of PM2.5.

Overall, the target-focused exploratory and stationarity analysis shows that Delhi PM2.5 is persistent, seasonally structured, right-skewed, and affected by irregular high-volatility episodes. These findings justify using SARIMA-family models, comparing both $d = 0$ and $d = 1$ specifications, considering seasonal terms with $s = 7$, and evaluating final candidates using out-of-sample performance together with residual diagnostics. The correlation and VIF results further motivate the selected exogenous variables used in the Log-SARIMAX extension.

Three | Methodology

This chapter describes the modeling workflow used to forecast daily PM2.5 concentration in Delhi. The methodological design follows a Box–Jenkins time-series procedure: the chronologically prepared training series is examined for temporal structure, several benchmark and SARIMA-family models are estimated, and the final models are evaluated through holdout testing, residual diagnostics, multi-horizon backtesting, and walk-forward validation. The purpose of this chapter is to explain the modeling logic and statistical tools used in the analysis.

BASELINE FORECASTING MODELS

Before estimating SARIMA, three baseline forecasting models are constructed to provide simple reference points for model evaluation. Baseline models are important because a more complex time-series model should only be considered useful if it can improve on transparent forecasting rules. In this project, the baselines are the naive forecast, the seasonal naive forecast, and Holt-Winters exponential smoothing.

The first benchmark is the naive forecast, which assumes that the most recent observed PM2.5 value contains the best available information about the immediate future. For every date in the test set, the forecast is equal to the final observation in the training set:

$$\hat{y}_{T+h} = y_T, \qquad h = 1, 2, \ldots, H,$$

where T is the last training date and H is the length of the test period. This benchmark is especially relevant for PM2.5 because air-pollution concentrations are persistent from one day to the next. If SARIMA cannot outperform this simple rule, then its additional complexity would not be justified.

The second benchmark is the seasonal naive forecast. Instead of repeating only the last value, this method repeats the most recent observed seasonal cycle from the training period. Because the data are recorded at daily frequency and the exploratory analysis shows strong annual pollution cycles, the seasonal period is set to $s = 365$. This means that the forecast for the test period is generated by reusing the last 365 daily PM2.5 observations from the training set and repeating this annual pattern as needed:

$$\hat{y}_{T+h} = y_{T-365+h}, \qquad h = 1, 2, \ldots, 365.$$

If the test horizon is shorter than 365 days, only the required portion of the repeated annual pattern is retained. This design is appropriate for Delhi PM2.5 because pollution levels tend to follow an annual seasonal rhythm, with higher concentrations during winter and lower concentrations during cleaner summer or monsoon periods.

The third benchmark is Holt-Winters exponential smoothing, also known as an ETS model. The implementation uses an additive trend and additive seasonality with `seasonal_periods=365`. The additive form is chosen because it provides an interpretable baseline in which the forecast is decomposed into a level component, a trend component, and an annual seasonal component. In general form, the forecast can be written as

$$\hat{y}_{T+h} = \ell_T + hb_T + s_{T-365+h},$$

where ℓ_T is the estimated level, b_T is the estimated trend, and $s_{T-365+h}$ is the seasonal factor associated with the corresponding day in the annual cycle. Using 365 seasonal periods allows the model to capture recurring within-year pollution patterns while still remaining simpler and more transparent than SARIMA.

Together, these baselines represent increasing levels of structure: the naive model captures short-term persistence, the seasonal naive model captures annual recurrence, and Holt-Winters captures level, trend, and annual seasonality. The SARIMA-family models are therefore evaluated against these benchmarks to determine whether they provide meaningful forecasting gains beyond simple persistence and seasonal repetition.

SARIMA ORDER SELECTION AND MODEL SPECIFICATION

The final SARIMA-family specification is obtained through a two-stage order-selection strategy that combines automated candidate generation, explicit grid search, holdout forecasting performance, and residual diagnostics. This design is used because order selection in seasonal ARIMA models should not rely on a single criterion alone. Information criteria such as AIC and BIC evaluate in-sample likelihood penalized by model complexity, while the main objective of this project is out-of-sample prediction on the 2020 test set.

The first stage uses `auto_arima` with weekly seasonality, setting $m = 7$ because the observations are daily and the diagnostic analysis identifies short-cycle weekly dependence. The search allows non-seasonal and seasonal orders within compact bounds, including $p \leq 3$, $q \leq 2$, $P \leq 2$, $Q \leq 1$, $d \leq 1$, and $D \leq 1$. All valid fitted candidates are retained and then ranked by holdout RMSE, while MAE, MAPE, AIC, and BIC are reported as supporting evidence. In general, the automated search tends to favor relatively parsimonious specifications and, in several candidate models, places little emphasis on differencing terms. This is consistent with the stationarity tests, which suggest that the original PM2.5 series is already stationary in the stochastic sense.

The second stage uses an explicit SARIMA grid search. Unlike the stepwise automated search, the grid search evaluates a more controlled set of candidate structures by fixing $d = 1$, $D = 1$, and $s = 7$, while varying $p \in \{0, 1, 2, 3\}$, $q \in \{0, 1, 2\}$, $P \in \{0, 1\}$, and $Q \in \{0, 1\}$. This wider manual comparison is useful because visual diagnostics show strong persistence, seasonal dependence, and volatility clustering even though formal stationarity tests do not require differencing. Each candidate is evaluated on the test set, and a model is considered eligible only if its residuals pass the Ljung-Box white-noise check at seasonal lags. Thus, the selected order must satisfy both predictive and diagnostic criteria: low validation RMSE and no statistically significant residual autocorrelation.

The information criteria used for reference are

$$\mathrm{AIC} = -2\ell + 2k,$$

$$\mathrm{BIC} = -2\ell + k \log(n),$$

where ℓ is the log-likelihood, k is the number of estimated parameters, and n is the sample size. Lower values indicate a better in-sample trade-off between fit and complexity. However, in this workflow AIC and BIC are secondary criteria; the final choice is guided primarily by holdout RMSE and residual adequacy because the purpose of the model is forecasting rather than only in-sample description.

The selected univariate model is SARIMA$(3, 1, 1)(0, 1, 1)_7$. The weekly seasonal period is used for the SARIMA-family search because it is computationally feasible and directly supported by the short-lag autocorrelation diagnostics. Annual recurrence is still represented in the comparison through the seasonal naive and ETS baselines with exttttseasonal_periods=365. A general SARIMA model is written as SARIMA$(p, d, q)(P, D, Q)_s$, where (p, d, q) denotes the non-seasonal autoregressive, differencing, and moving-average orders, and $(P, D, Q)_s$ denotes the corresponding seasonal orders with seasonal period s. Using the backshift operator B, the model can be expressed as

$$\Phi_P(B^s)\phi_p(B)(1 - B)^d(1 - B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t,$$

where $\phi_p(B)$ and $\theta_q(B)$ are the non-seasonal autoregressive and moving-average polynomials, $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are the seasonal autoregressive and moving-average polynomials, and ε_t is a white-noise innovation. For the selected model, $p = 3$, $d = 1$, $q = 1$, $P = 0$, $D = 1$, $Q = 1$, and $s = 7$. This means that the model combines three short-run autoregressive lags, one ordinary difference, one non-seasonal moving-average term, weekly seasonal differencing, and one seasonal moving-average term.

In addition to the univariate SARIMA model, the workflow estimates a Log-SARIMAX model. This extension is motivated by two empirical features of the dataset. First, PM2.5 has non-constant variance, with larger fluctuations during high-pollution episodes. Second, PM2.5 is correlated with other pollutants, so exogenous information may improve forecasting performance. Benzene, NOx, and NH3 are used as exogenous regressors because they are related to PM2.5 and are selected after correlation and multicollinearity screening. Because SARIMAX requires future values of the exogenous regressors when forecasting, the holdout comparison treats these pollutant variables as observed contemporaneous inputs over the evaluation period. The resulting Log-SARIMAX results should therefore be interpreted as conditional forecasts that measure the value of additional pollutant information, not as a fully operational system that forecasts all predictors independently.

The Log-SARIMAX target transformation is

$$z_t = \log(1 + y_t),$$

and each exogenous predictor is transformed analogously as $\log(1 + x_{jt})$. The use of `log1p` is appropriate because it stabilizes variance while remaining well-defined for zero-valued observations. The model can be written as

$$z_t = \boldsymbol{x}_t' \boldsymbol{\beta} + u_t,$$

where $\boldsymbol{x}_t$ is the vector of transformed exogenous variables and u_t follows the same SARIMA error structure:

$$\Phi_P(B^s)\phi_p(B)(1 - B)^d(1 - B^s)^D u_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t.$$

Thus, Log-SARIMAX combines regression on external pollution indicators with a seasonal ARIMA process for the remaining serial dependence. Forecasts are converted back to the original PM2.5 scale using

$$\hat{y}_t = \exp(\hat{z}_t) - 1.$$

This back-transformation allows SARIMA and Log-SARIMAX to be compared using the same MAE, RMSE, and MAPE metrics on the original μg/m^3 scale.

Overall, the order-selection results show that automated and manual procedures provide complementary evidence. Auto-ARIMA is useful for generating parsimonious candidates and confirming that low-differencing models are plausible, while grid search more deliberately explores differenced seasonal structures and imposes residual white-noise checks. The final SARIMA$(3, 1, 1)(0, 1, 1)_7$ specification is therefore not chosen solely from theoretical ACF/PACF rules or from one information criterion, but from a combined assessment of diagnostics, validation performance, and residual behavior. The Log-SARIMAX extension then tests whether adding variance stabilization and exogenous pollutant information improves forecast accuracy beyond the univariate seasonal model.

FORECAST EVALUATION METRICS

Forecast accuracy is evaluated using MAE, RMSE, and MAPE. For test observations y_t and forecasts $\hat{y}_t$, Mean Absolute Error is

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t|.$$

Root Mean Squared Error is

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}.$$

Mean Absolute Percentage Error is

$$\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|.$$

MAE is interpretable in μg/m^3, RMSE penalizes large errors more strongly, and MAPE expresses error as a percentage of the observed value. Because PM2.5 values are positive, MAPE can be interpreted meaningfully, although it may become large when actual values are relatively low.

RESIDUAL DIAGNOSTICS

Residual diagnostics are used to evaluate whether the fitted model has captured the main temporal structure. The residual is

$$e_t = y_t - \hat{y}_t.$$

A satisfactory time-series model should have residuals centered near zero, stable variance, and no significant autocorrelation.

The Ljung-Box test is used to test residual white noise. Its statistic is

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{n-k},$$

where $\hat{\rho}_k$ is the residual autocorrelation at lag k, h is the maximum lag, and n is the sample size. The null hypothesis is that residual autocorrelations up to lag h are jointly zero. Lags 1, 7, and 14 are checked so that both immediate and weekly residual dependence are assessed.

MULTI-HORIZON AND WALK-FORWARD VALIDATION

Beyond the fixed holdout forecast, the workflow evaluates model performance under multi-horizon backtesting. This is necessary because forecast accuracy usually depends on how far ahead the model is asked to predict. A one-day-ahead forecast can rely heavily on the most recent pollution level, whereas three-day and seven-day forecasts must project the time-series dynamics further into the future and are therefore more exposed to accumulated uncertainty.

Forecast horizons of 1, 3, and 7 days are considered. For a given forecast origin t and horizon h, the model produces a prediction $\hat{y}_{t+h|t}$ using only information available up to time t. This prediction is then compared with the realized value y_{t+h}. The forecast error at horizon h is therefore

$$e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}.$$

Repeating this procedure across the test period produces a set of horizon-specific forecast errors, which are summarized using MAE, RMSE, and MAPE. Comparing the metrics across $h = 1$, $h = 3$, and $h = 7$ shows whether the model is mainly useful for immediate short-term forecasting or whether it remains reliable over a full weekly horizon.

The most realistic validation design is walk-forward validation. In a standard fixed holdout forecast, the model is estimated once on the training set and then forecasts the entire test period without incorporating newly observed test values. In contrast, walk-forward validation mimics an operational forecasting environment. At each step, the model first forecasts the next observation;

after the true value becomes available, that observation is appended to the model state before the next forecast is generated. Thus, each prediction uses the maximum information that would have been available at that date, while still preventing future data from entering the forecast origin.

Walk-forward validation is implemented by fitting the SARIMA model on the training set, generating a one-step-ahead forecast for the next test date, and then updating the fitted state with the observed value using `append(..., refit=False)`. This update changes the information set without re-estimating all model parameters at every step, making the procedure computationally efficient while still reflecting real-time forecast updating. The same logic is also applied in the multi-horizon loop for SARIMA and Log-SARIMAX, where the model state is advanced through the test sequence and exogenous variables are supplied for the required forecast horizon.

This validation strategy provides a stronger assessment than a single static forecast because it separates three questions: how accurate the model is at different horizons, how well it adapts when new observations arrive, and whether the selected SARIMA-family structure remains stable throughout the 2020 test period. For this reason, the walk-forward results are treated as the most operationally relevant evidence of short-term forecasting performance.

Four | Results and Analysis

This chapter presents the forecasting results for the holdout period from 1 January 2020 to 1 July 2020. The discussion is organized around four questions: whether the individual models improve on simple benchmarks, whether the selected SARIMA order is statistically adequate, whether Log-SARIMAX adds useful information beyond the univariate model, and where the forecast errors are largest. The same MAE, RMSE, and MAPE criteria are used throughout, so the models are compared on a common PM2.5 scale. Lower MAE and RMSE indicate smaller absolute forecast errors in μg/m^3, while lower MAPE indicates smaller relative errors.

The order of presentation is deliberate. The chapter first compares forecast accuracy on the fixed holdout set because this is the most direct out-of-sample comparison across candidate models. It then examines the selected SARIMA parameters and residual diagnostics to determine whether weak forecast accuracy is caused by an inadequate temporal specification or by the difficulty of long-horizon static prediction. Finally, it evaluates walk-forward and multi-horizon forecasts, which are closer to a real daily forecasting workflow.

HOLDOUT PERFORMANCE OF INDIVIDUAL MODELS

The baseline results show that model structure matters. The naive forecast performs poorly because it carries the last training value forward across the entire 183-day test period. This produces MAE of 167.487 μg/m^3, RMSE of 174.364 μg/m^3, and MAPE of 335.259%. The result is expected because Delhi PM2.5 changes strongly between winter, spring, and early summer; a single December 2019 value cannot represent the full January–July 2020 evaluation window. The seasonal naive forecast performs much better, reducing RMSE to 58.246 μg/m^3, because it reuses the annual pollution pattern from the previous year. This confirms that annual seasonality is an important feature of the Delhi PM2.5 series. Holt-Winters ETS is the strongest baseline, with RMSE of 39.340 μg/m^3 and MAE of 30.493 μg/m^3, because it combines level, trend, and annual seasonal components rather than relying only on persistence or repetition.

Table 1: Baseline Forecast Accuracy on the Holdout Test Set

Model	MAE	RMSE	MAPE (%)
Naive	167.487	174.364	335.259
Seasonal naive	44.062	58.246	75.413
Holt-Winters ETS	30.493	39.340	52.118

The static SARIMA forecast does not perform well when the model is estimated once and then used to forecast the whole holdout period without state updating. SARIMA$(3,1,1)(0,1,1)_7$ records RMSE of 163.137 μg/m^3, which is close to the naive benchmark and much worse than ETS. This does not mean that the SARIMA structure is useless; rather, it shows that a long static forecast is difficult for a model built around short-run dependence and weekly seasonal adjustment. The model uses $s = 7$ to capture short-cycle dependence, while the fixed holdout period spans a seasonal transition from winter to early summer. Once the forecast origin moves far away from the training sample, errors can accumulate and the model may remain anchored to winter pollution levels even as actual PM2.5 declines.

Log-SARIMAX substantially improves over the static SARIMA forecast. Its RMSE is 73.405 μg/m^3, compared with 163.137 μg/m^3 for SARIMA, equivalent to a 55.00% reduction in RMSE. MAE also falls from 155.643 to 71.431 μg/m^3. The improvement has two likely causes. First, the logarithmic transformation reduces the influence of extreme PM2.5 spikes and stabilizes the variance. Second, Benzene, NOx, and NH3 provide contemporaneous pollutant information that helps explain daily PM2.5 movements that cannot be captured from PM2.5 history alone. However, Log-SARIMAX is still less accurate than ETS in the fixed holdout comparison, so the benefit of exogenous information does not fully offset the difficulty of forecasting the full test period statically.

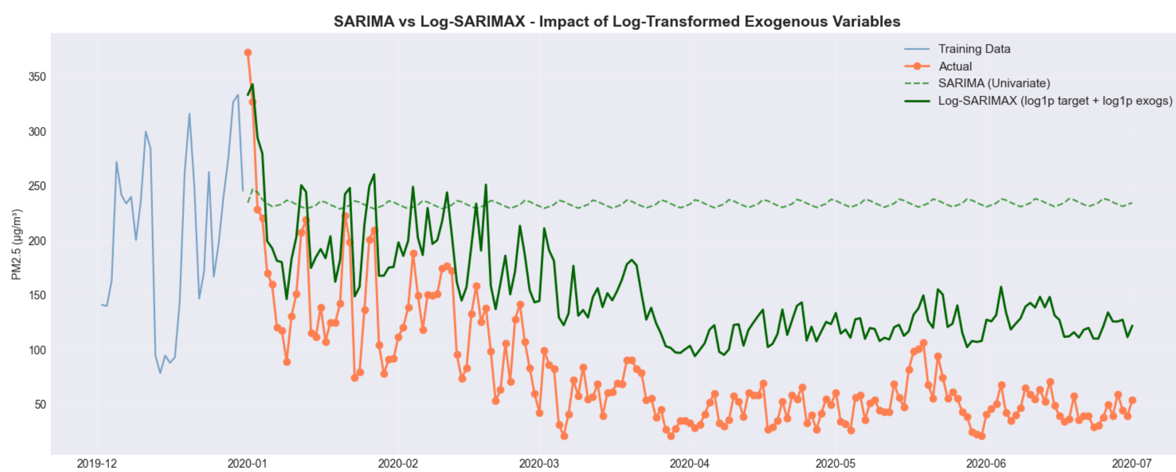

Figure 5: Static holdout forecasts from SARIMA and Log-SARIMAX compared with observed PM2.5.

Table 2: Static SARIMA-Family Forecast Accuracy on the Holdout Test Set

Model	MAE	RMSE	MAPE (%)
$\text{SARIMA}(3,1,1)(0,1,1)_7$	155.643	163.137	314.422
Log-SARIMAX	71.431	73.405	133.377

SELECTED SARIMA ORDER AND MODEL SUMMARY

The selected univariate specification is $\text{SARIMA}(3,1,1)(0,1,1)_7$. This subsection is placed after the holdout comparison because it explains why the SARIMA model is statistically reasonable even though its static long-horizon forecast is weak. The non-seasonal AR terms show that daily PM2.5 has short-run memory: the first autoregressive coefficient is positive and large ($\hat{\phi}_1 = 0.7315$), while the second coefficient is negative ($\hat{\phi}_2 = -0.1874$) and the third coefficient is positive ($\hat{\phi}_3 = 0.1147$). All three are statistically significant with p-values below 0.01. This pattern suggests that PM2.5 is highly persistent from one day to the next, but the influence of earlier lags is corrective rather than purely monotonic. The non-seasonal moving-average coefficient is also large and negative ($\hat{\theta}_1 = -0.9078$), indicating that recent shocks are strongly absorbed by the error-correction component of the model. The seasonal moving-average coefficient at lag 7 is nearly -1 and statistically significant, which supports the inclusion of weekly seasonal error structure.

The in-sample information criteria for the fitted SARIMA model are AIC = 18612.439 and BIC = 18645.472. These values are mainly used for model comparison within the SARIMA class, not as direct measures of forecast accuracy. The model summary also reports residual variance $\hat{\sigma}^2 = 1589.7234$, which is large on the original PM2.5 scale and is consistent with the high volatility observed in the raw series. The Jarque-Bera test strongly rejects normality, with skewness of 1.02 and kurtosis of 14.01. Therefore, even though the model captures serial dependence, the residual distribution remains heavy-tailed. This is reasonable for Delhi PM2.5 because pollution episodes can generate abrupt spikes that are difficult for a Gaussian linear time-series model to represent.

Table 3: Selected SARIMA Model Summary

Term	Coefficient	Std. Error	z-statistic	p-value
AR(1)	0.7315	0.021	34.406	0.000
AR(2)	-0.1874	0.017	-10.825	0.000
AR(3)	0.1147	0.019	5.909	0.000
MA(1)	-0.9078	0.016	-58.057	0.000
Seasonal MA(7)	-0.9997	0.133	-7.522	0.000

ROLE OF EXOGENOUS POLLUTANTS IN LOG-SARIMAX

The Log-SARIMAX coefficient estimates help explain why the model improves over static SARIMA. Benzene, NOx, and NH3 all have positive and statistically significant coefficients in the log-transformed equation. Benzene has the largest coefficient (0.3941), followed by NH3 (0.2854) and NOx (0.1502). Because the model is estimated in log form, these coefficients can be interpreted approximately as elasticities, holding the SARIMA error process constant. A higher value of these pollutants is associated with higher predicted PM2.5. This is substantively reasonable because these variables reflect combustion, traffic, atmospheric chemistry, and other pollution conditions that move together with fine particulate matter.

The time-series terms remain significant even after adding exogenous predictors. The Log-SARIMAX AR(1) coefficient is 0.6297, and the seasonal MA(7) coefficient is -0.9986, both with p-values below 0.001. This indicates that exogenous pollutants do not replace temporal structure; instead, they complement it. The model benefits from both sources of information: contemporaneous pollutant covariation and historical PM2.5 dependence. The conditional nature of this model should also be emphasized. Since future values of Benzene, NOx, and NH3 are supplied for the holdout period, the Log-SARIMAX forecasts should be interpreted as conditional forecasts that measure the value of additional pollutant information, not as a standalone system that independently forecasts every predictor.

Table 4: Log-SARIMAX Key Coefficients

Term	Coefficient	Std. Error	z-statistic	p-value
Benzene	0.3941	0.019	20.304	0.000
NOx	0.1502	0.019	7.946	0.000
NH3	0.2854	0.027	10.620	0.000
AR(1)	0.6297	0.026	23.941	0.000
Seasonal MA(7)	-0.9986	0.030	-33.774	0.000

RESIDUAL ADEQUACY AND REMAINING LIMITATIONS

Residual autocorrelation diagnostics are satisfactory for both SARIMA-family models. For SARIMA, the Ljung-Box p-values are 0.6240 at lag 7, 0.2299 at lag 14, and 0.2448 at lag 21. For Log-SARIMAX, the corresponding p-values are 0.0777, 0.1937, and 0.2114. Since all p-values exceed 0.05, the null hypothesis of no residual autocorrelation is not rejected at the weekly lags tested. This means the models have largely removed the remaining linear serial dependence after accounting for autoregressive, moving-average, differencing, seasonal, and exogenous structures.

The residual statistics provide a more nuanced interpretation. The SARIMA residual mean is 1.0836, which is close to zero relative to the scale of PM2.5, so the model is not strongly biased on average. However, the residual standard deviation is 41.245, with a minimum of -242.850 and a maximum of 313.220. These wide residual bounds show that some pollution episodes remain difficult to predict even when autocorrelation has been removed. The SARIMA residual distribution is also right-skewed, with skewness of 1.127, and heavy-tailed, with kurtosis of 11.386. Therefore, SARIMA captures temporal dependence adequately, but it does not fully capture extreme shocks or non-normal residual behavior.

Log-SARIMAX improves several residual properties. Its in-sample residual mean is 0.0038, indicating almost no average bias on the log scale, and the residual standard deviation is approximately 0.246. The Shapiro-Wilk test still rejects normality, and 84 observations, or 4.60% of the training sample, are flagged as outliers. The variance-stability ratio is 0.759, which is classified as weak rather than ideal. Nevertheless, the Log-SARIMAX residuals pass the white-noise check at lags 7, 14, and 21. This supports the conclusion that the model has captured the main temporal structure, while the remaining errors are mainly related to outliers, changing volatility, and extreme pollution events rather than systematic autocorrelation.

Table 5: Residual White-Noise Diagnostics

Model	LB(7)	LB(14)	LB(21)	Conclusion
SARIMA	0.6240	0.2299	0.2448	White noise
Log-SARIMAX	0.0777	0.1937	0.2114	White noise

Table 6: Residual Distribution Summary

Model	Mean	Std. Dev.	Minimum	Maximum	Skewness	Kurtosis
SARIMA	1.0836	41.245	-242.850	313.220	1.127	11.386
Log-SARIMAX	0.0038	0.246	–	–	–	–

Log-SARIMAX diagnostics also identify 84 outliers, equal to 4.60% of the training sample, and a variance-stability ratio of 0.759. Dashes indicate that the corresponding distributional statistics were not reported for the log-scale residual output used in this table.

WALK-FORWARD AND MULTI-HORIZON FORECASTING

The walk-forward validation results are much more favorable for SARIMA than the static holdout results. When the model is updated after each observed test value, SARIMA achieves RMSE of 27.824 μg/m^3, MAE of 19.121 μg/m^3, and MAPE of 27.333%. This large improvement shows that SARIMA is primarily useful as a short-term adaptive forecasting model. The model performs poorly when it must forecast many months ahead without new information, but it performs well when the information set is refreshed daily.

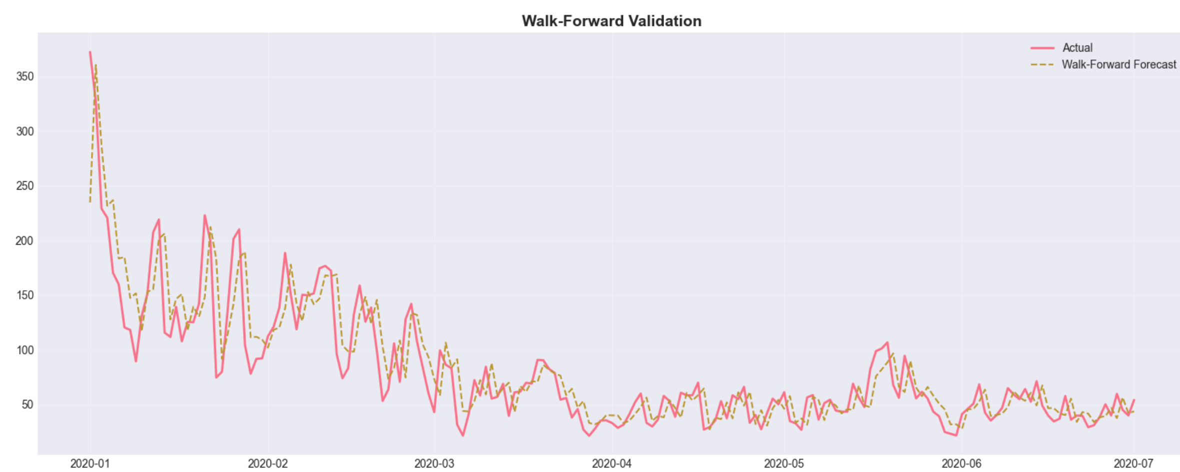

Figure 6: Walk-forward SARIMA forecasts compared with observed PM2.5 in the test period.

The multi-horizon results show a clear horizon effect. SARIMA RMSE increases from 27.824 at the one-day horizon to 37.880 at the three-day horizon and 39.891 at the seven-day horizon. Log-SARIMAX also becomes less accurate as the horizon increases, but its errors remain lower than SARIMA at every horizon. At the one-day horizon, Log-SARIMAX reduces RMSE from 27.824 to 18.184 μg/m^3. At the seven-day horizon, Log-SARIMAX RMSE is 26.141 μg/m^3, still below the one-day SARIMA RMSE. This result suggests that exogenous pollutant information and variance stabilization are especially valuable for short-run PM2.5 prediction.

Table 7: Multi-Horizon Backtesting Accuracy

Horizon	S-RMSE	S-MAE	S-MAPE	LS-RMSE	LS-MAE	LS-MAPE
1 day	27.824	19.121	27.333	18.184	13.375	20.103
3 days	37.880	26.844	40.695	25.493	19.023	29.179
7 days	39.891	27.984	45.635	26.141	19.616	33.338

Notes: S = SARIMA; LS = Log-SARIMAX.

FORECAST ERROR INTERPRETATION

The forecast errors are smallest when pollution evolves gradually and recent observations remain informative. This is why walk-forward SARIMA performs well: each new actual PM2.5 value updates the model state and prevents the forecast path from drifting too far from current pollution conditions. Log-SARIMAX performs particularly well when the exogenous variables move consistently with PM2.5. For example, on 1 January 2020, actual PM2.5 is 372.14 $\mu g/m^3$, and the one-day Log-SARIMAX forecast is 374.000 $\mu g/m^3$, showing that the model can track high-pollution conditions when the contemporaneous pollutant signals are informative.

Errors increase when PM2.5 changes abruptly or when the forecast horizon is extended. Early January examples show that SARIMA seven-day forecasts can remain too high when actual PM2.5 falls quickly from severe winter levels toward lower values. This reflects a common limitation of autoregressive models: persistence is useful when the process changes slowly, but it can delay adjustment after a sudden regime shift. MAPE is especially sensitive in this setting because percentage errors become large when actual PM2.5 values are low. Therefore, RMSE and MAE should be treated as the primary accuracy measures, while MAPE is interpreted as supplementary evidence of relative error.

Overall, the structure of the evidence is consistent. Fixed holdout results answer which model works best when the forecast path is generated without updating; diagnostics explain whether the fitted SARIMA-family models leave systematic autocorrelation; and walk-forward results show how performance changes in a more operational setting. The results support three conclusions. First, annual seasonal baselines such as ETS are strong competitors in fixed holdout forecasting because they capture broad winter–summer movement. Second, SARIMA residual diagnostics are statistically adequate, but static multi-month SARIMA forecasts are weak because errors accumulate when the model is not updated. Third, Log-SARIMAX provides the strongest short-horizon performance because it combines temporal dependence, weekly seasonal error correction, variance stabilization, and additional pollutant information. For operational daily air-quality forecasting, the most defensible approach is therefore an updated short-horizon SARIMA-family workflow, with Log-SARIMAX preferred when reliable exogenous pollutant inputs are available.

Five | Discussion

The results show that PM2.5 forecasting in Delhi benefits from models that recognize both temporal structure and changing pollution intensity. Daily PM2.5 is not a random sequence: it contains persistence from one day to the next, weekly dependence, and broader seasonal movement associated with winter pollution episodes and cleaner monsoon or summer periods. A model that ignores these patterns can easily overreact to one recent observation or fail to follow recurring pollution cycles. This explains why the naive benchmark performs poorly, while seasonal and time-series models provide more meaningful forecasts.

Seasonal structure improves forecasting because Delhi PM2.5 follows repeated patterns created by meteorology, emissions, and atmospheric accumulation. Winter months often have higher pollution because lower mixing height and stagnant air trap pollutants close to the ground, while warmer and monsoon periods usually show lower concentrations. In the modeling workflow, this seasonality is represented in different ways. The seasonal naive and ETS baselines use annual recurrence, while the SARIMA-family models use weekly seasonal differencing and seasonal moving-average terms to capture shorter-cycle dependence. The results suggest that both forms of seasonality are useful, but they serve different purposes. Annual seasonality helps explain broad level shifts across the fixed holdout period, whereas weekly SARIMA structure is more useful for short-horizon adaptive forecasting.

The logarithmic transformation also improves forecasting because the original PM2.5 series is right-skewed and heteroscedastic. High-pollution days create large spikes, and the forecast errors on those days can dominate RMSE. By modeling $\log(1 + y_t)$ rather than raw PM2.5, Log-SARIMAX compresses extreme values and makes the variance more stable. This is supported by the residual diagnostics: compared with the raw-scale SARIMA model, Log-SARIMAX shows a much smaller residual scale and weaker evidence of heteroscedasticity. The transformation therefore makes the model less sensitive to extreme pollution episodes while still allowing forecasts to be converted back to the original μg/m^3 scale for interpretation.

From a practical perspective, the strongest value of the SARIMA-family approach is short-term air-quality monitoring rather than long-range static prediction. The static SARIMA forecast performs poorly when asked to project the full January–July 2020 test period without updating, because errors accumulate as the forecast origin moves further from the training sample. In contrast, walk-forward validation produces much lower errors because each new observed PM2.5 value updates the model state before the next forecast is made. This resembles a realistic monitoring system: agencies observe today's pollution, update the forecasting model, and issue tomorrow's warning. Under this setting, SARIMA becomes useful for daily public-health alerts, school and workplace advisories, and short-run pollution-management decisions.

The comparison between baseline SARIMA and Log-SARIMAX highlights the benefit of adding exogenous pollutant information. SARIMA is transparent and relies only on historical PM2.5, so it is easy to implement and useful when no other pollutant forecasts are available. Its residual white-noise diagnostics also indicate that the main serial dependence has been captured. However, SARIMA cannot directly use information from related pollutants, and its raw-scale residuals remain heavy-tailed. Log-SARIMAX addresses these weaknesses by combining log transformation with Benzene, NOx, and NH3 as exogenous regressors. These variables are statistically significant and positively associated with PM2.5, suggesting that they provide useful information about broader pollution conditions.

Overall, SARIMA should be interpreted as a strong and interpretable univariate benchmark, while Log-SARIMAX is the more informative extension when reliable exogenous pollutant inputs are available. The baseline SARIMA model answers how much can be learned from PM2.5 history alone. The Log-SARIMAX model answers whether forecast accuracy improves when the model also uses contemporaneous pollutant covariation and variance stabilization. The empirical results favor Log-SARIMAX for short-horizon forecasting, especially in the multi-horizon backtesting results, but this advantage depends on having valid future or real-time exogenous inputs. Therefore, the most practical conclusion is not that one model universally dominates, but that model choice should depend on the forecasting environment: SARIMA is appropriate for simple univariate monitoring, while Log-SARIMAX is preferable for richer operational systems with additional pollutant data.

Six | Limitations and Future Work

This study provides a structured comparison of baseline forecasts, SARIMA, and Log-SARIMAX for daily PM2.5 concentration in Delhi. However, several limitations should be considered when interpreting the results. These limitations do not invalidate the analysis, but they define the scope within which the findings are most reliable.

The first limitation is that the analysis uses a single-city daily time series. Focusing on Delhi makes the modeling objective clear and avoids mixing observations from cities with different pollution sources, meteorology, and policy environments. At the same time, this design limits external validity. The selected SARIMA order and the relative performance of Log-SARIMAX may not generalize to other Indian cities, where seasonal cycles, emission profiles, and pollutant relationships can differ substantially. A broader study would need to estimate and compare models across multiple cities or use a panel-style forecasting design.

The second limitation is the relatively short holdout horizon. The test period covers 1 January 2020 to 1 July 2020, which provides 183 daily observations for out-of-sample evaluation. This period is useful for evaluating short-run forecasting behavior, but it does not cover a full annual cycle. As a result, the holdout set may not fully represent all seasonal conditions, especially the late-autumn and winter pollution episodes that are central to Delhi's air-quality problem. A longer evaluation window or rolling-origin backtesting across several years would provide a stronger assessment of model stability.

The third limitation is sensitivity to extreme pollution spikes. Delhi PM2.5 contains abrupt high-pollution events caused by meteorological stagnation, regional emissions, dust, combustion activity, and other short-lived shocks. These events generate heavy-tailed residuals and can strongly affect RMSE and MAPE. The logarithmic transformation in Log-SARIMAX helps reduce this problem by compressing extreme values, but it does not eliminate it completely. Forecasting severe pollution episodes may require additional explanatory variables and models designed to handle nonlinear or threshold behavior.

Another limitation is that the current code does not implement an ensemble forecasting pipeline. Ensemble forecasting could combine SARIMA, ETS, Log-SARIMAX, and possibly machine-learning forecasts to reduce reliance on a single model. This is relevant because different models perform well under different conditions: ETS captures annual seasonal movement, SARIMA performs well in walk-forward short-horizon updating, and Log-SARIMAX benefits from exogenous pollutant information. A weighted or stacked ensemble could potentially produce more robust forecasts, but such a system is outside the scope of the present analysis.

Future work should extend the model in three main directions. First, meteorological variables should be added, including temperature, humidity, wind speed, wind direction, rainfall, boundary-layer height, and atmospheric pressure [10, 11]. These variables are directly related to pollutant dispersion and accumulation, so they may help explain sudden changes in PM2.5 that are difficult to infer from pollutant history alone. Second, richer multivariate specifications should be tested [9, 10, 11]. Vector autoregression, dynamic regression, SARIMAX with additional pollutant and weather inputs, or regularized multivariate models could better capture interactions among PM2.5, PM10, NOx, NH3, Benzene, and meteorological conditions.

Third, future research could compare classical time-series models with deep-learning approaches [4, 12]. Models such as LSTM, GRU, temporal convolutional networks, or hybrid SARIMA–neural network models may capture nonlinear temporal patterns that SARIMA cannot represent [4, 12]. However, these models should be evaluated carefully against transparent baselines, because greater complexity does not automatically imply better out-of-sample performance. The most useful extension would therefore combine rigorous rolling-origin validation, interpretable benchmark models, exogenous meteorological variables, and more flexible nonlinear methods.

Overall, the current analysis should be viewed as a transparent forecasting benchmark rather than a final operational forecasting system. It shows that SARIMA-family models can capture important temporal structure in Delhi PM2.5 and that Log-SARIMAX improves short-horizon accuracy when exogenous pollutant inputs are available. Future work can build on this foundation by widening the data sources, lengthening the evaluation period, and testing ensemble, monitoring-system, or deep-learning models under the same strict out-of-sample validation framework [4, 7, 12].

Conclusion

This study examined daily PM2.5 forecasting in Delhi using a sequence of benchmark and SARIMA-family models. The analysis began with simple baseline forecasts, then estimated a univariate SARIMA model, and finally extended the framework to Log-SARIMAX with log-transformed exogenous pollutant variables. All models were evaluated on the same chronological holdout period, with additional walk-forward and multi-horizon validation used to assess operational short-term performance.

The model comparison shows that simple seasonal structure is important. The naive forecast performs poorly because it cannot adapt to changing pollution levels across the test period, while the seasonal naive and Holt-Winters ETS baselines perform much better by capturing annual recurrence and level movement. The static SARIMA forecast is weak over the full holdout horizon, mainly because a model estimated once on the training period accumulates errors when forecasting several months ahead without updating. However, residual diagnostics show that the selected $\mathrm{SARIMA}(3,1,1)(0,1,1)_7$ model captures the main temporal dependence: Ljung-Box tests at weekly lags do not indicate significant remaining autocorrelation.

The validation results also show that SARIMA is much more useful in a walk-forward setting. When the model is updated sequentially as new PM2.5 observations become available, forecast accuracy improves substantially. This finding is important because daily air-quality forecasting is normally an updating problem rather than a one-time multi-month prediction problem. The Log-

SARIMAX extension gives the strongest short-horizon performance because it combines temporal dependence, weekly seasonal structure, variance stabilization, and exogenous pollutant information from Benzene, NOx, and NH3. Its lower multi-horizon errors indicate that additional pollutant information helps explain PM2.5 movements that cannot be captured by PM2.5 history alone.

The final recommendation is therefore to use SARIMA-family models as transparent short-term forecasting tools, but to prefer Log-SARIMAX when reliable exogenous pollutant inputs are available. For a simple monitoring system with only historical PM2.5 data, walk-forward SARIMA is an interpretable and practical baseline. For a richer operational system that can observe or forecast related pollutants, Log-SARIMAX is recommended because it provides better short-horizon accuracy and more stable residual behavior. In both cases, the model should be updated regularly and evaluated with rolling or walk-forward validation rather than judged only by a static long-horizon forecast.

Overall, the project shows that daily PM2.5 forecasting in Delhi benefits from combining seasonal time-series structure with careful validation. SARIMA provides a useful statistical benchmark, while Log-SARIMAX offers a stronger practical forecasting specification when additional pollutant information is available. Future improvements should extend the data with meteorological variables, test ensemble and deep-learning methods, and evaluate performance over longer periods and additional cities.

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